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RESEARCH

Physics-Informed neural SOH Estimation method for Lithium-ion battery under partial observability and sparse sensor data

Ma Jin[1] · Xu Ming[1] · Du Wei[1] · Qiao Zhongzhi[1] · Zhang Jiahui[2] · Bai Xiang[2] · Qiu Pengfei[2] · Wang Lei[2]

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Abstract

Establishment of precise state-of-health (SOH) consists of a vital step to guarantee reliable, secure, and preventive maintenance of electric vehicle (EV) fleets and grid-associated energy storage systems based on lithium-ion batteries. Nevertheless, existing SOH estimation approaches primarily consider situations where complete sensor profiles or full charge/discharge conditions are available, which is not often the case due to sensor degradation, communication losses, or economic factors. This work introduces a novel Physics-Informed Neural Network (PINN) framework to enable robust SOH estimation under partial observability, leveraging minimal sensor data and physics-constrained learning to ensure both accuracy and interpretability. The model relies on an approximation of the PINN architecture that is able to embed the fundamental degradation mechanisms, such as solid electrolyte interphase (SEI) layer growth and capacity fading, as information that the training process can utilize through physical regularization. While PINN is semantically mixed with traditional purely data-driven models, it uses a composite loss function that leverages data consistency with underlying electrochemical laws. This enables accurate predictions even when no further capacity measurements or complete voltage profiles are available. The framework is tested in the case of publicly available lithium-ion battery data sets, which show superior generalization and the capacity to withstand situations where missing inputs or sensor noise are introduced under simulated sensor sparsity scenarios. The results of the experiments demonstrate the quality of the model, which enhances the precision of SOH estimation in conditions of partial observability and also increases the model's interpretability and applicability in real-time in various operational settings. Statistical results show that the proposed method achieves a mean absolute error (MAE) of 0.008, root mean square error (RMSE) of 0.011, and a coefficient of determination (R^2) of 0.96, outperforming baseline models such as CNN–LSTM and hybrid SE-NN by 25–40% in sparse data regimes. Furthermore, the framework exhibits high generalization capability across different chemistries and retains robustness with as little as 50% of the input features. These results underscore the practical potential of the proposed PINN approach for real-time, physically consistent battery health diagnostics in embedded battery management systems.

Keywords Lithium-ion battery · SOH estimation · Physics-informed neural network · Hybrid modeling · Partial observability · Sparse input data

Nomenclatures

SOH	State of Health
PINN	Physics-Informed Neural Network
SEI	Solid Electrolyte Interphase
EV	Electric Vehicle
LIBs	Lithium-Ion Batteries
BMS	Battery Management System
IC	Incremental Capacity
EIS	Electrochemical Impedance Spectroscopy
PCA	Principal Component Analysis
NCA	Lithium Nickel Cobalt Aluminum Oxide
NCM	Lithium Nickel Cobalt Manganese Oxide
PENN	Physically Enhanced Neural Network
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
CNN	Convolutional Neural Network

✉ Wang Lei
253107842@qq.com

[1] Shanxi State Grid Changzhi Electric Power Supply Company, Changzhi, Shanxi 046000, China

[2] Shanxi Energy Internet Research Institute, Taiyuan, Shanxi 030000, China

LSTM	Long Short-Term Memory
SE-NN	Semi-Empirical Neural Network
ML	Machine Learning
GRU	Gated Recurrent Unit
TCN	Temporal Convolutional Network
SVR	Support Vector Regression
GPR	Gaussian Process Regression
GNN	Graph Neural Network
R^2	Coefficient of determination

Introduction

Research background and motivation

The emerging trend and recent growth in the field of EVs and energy storage systems, driven by LIBs and SOH assessment, have become a significant factor in the efficient and safe operation of such systems, as well as their reliability when used in long-haul operations. The issues associated with traditional SOH estimation methods, which are generally based on either physics models or machine learning, are becoming more pronounced due to the augmented requirements for on-demand operation of battery systems in variable and often non-optimal conditions. Both physics models and machine learning struggle with scalability, generalizability, and interpretability. The implementation of experimentally based methods that guarantee physical consistency and mechanism interpretability is, however, a challenge due to the high computational cost and the complicated nature of the extensive parameters involved.

On the other hand, machine-learning algorithms have the advantage of speedy and flexible estimation, but at the same time, they place too much stress on the quality and quantity of available data. This increased awareness of the inherent trade-offs led to additional attempts to create hybrid approaches, which combine the model interpretability of physics-based approaches with the learning capability of data-driven methods. Among all, Physics-Informed Neural Networks (PINNs) have become very popular as a way to penalize the absence of a complete set of physical information in the training data and thereby reduce the complexity of the problem. This feature can play a vital role in both theory and practice, especially in cases where actual data carrying full voltage profiles, temperature data, or historical cycling information is not accessible. Hence, it is critical to have SOH estimation frameworks that are not only data-efficient but also physically consistent, and achieve this goal, especially under partial observability.

In some situations, conducting battery observations poses a risk of being only partially observable, where important pieces of information, such as full voltage profiles or temperature traces, may not be available. This necessitates the development of SOH estimation models that are scalable and reliable, even under the most challenging conditions. Without a doubt, the urgency and significance of such a hybrid approach, which considers not only physical consistency but also data efficiency, becomes the focus and gives rise to the research motivation. Through our research, we aim to fill this gap by presenting a physics-based learning framework that enables high accuracy while maintaining the interpretability of the model, even with limited operational data available. Through the creation of machine models, it aims to make a difference and develop a new generation of battery management systems that not only provide reliable and efficient operation but are also scalable for multiple battery chemistries and usage profiles. Figure 1. Conceptual overview of battery SOH modeling challenges. Traditional models are physics-based, hence interpretable but rigid, while their successors are data-driven, implying obscurity but flexibility. Partial observability of sensor data further complicates accurate SOH estimation in practical BMS applications. This motivates the proposed physics-constrained hybrid learning framework.

Literature survey

The state of health estimation of LIBs plays a critical role in advancing EVs, large-scale energy storage systems, and battery second-life applications. As the demand for reliable and intelligent BMSs increases, the research focus has shifted toward developing more robust, interpretable, and computationally feasible SOH estimation methods. This section reviews the evolution of SOH estimation techniques, beginning with foundational review papers that define the landscape, followed by physics-based models, data-driven approaches, hybrid frameworks, and recent advances in physics-informed neural networks (PINNs), with a focus on their performance under partial observability and real-world constraints.

The review paper [1] provides a systematic classification of SOH estimation frameworks by introducing serial, parallel, and hybrid configurations that integrate physics-based and machine learning (ML) methods. It emphasizes the importance of leveraging both physical interpretability and data-driven adaptability. In a related context, the article [2] highlights the trend toward digitalizing lithium-ion batteries through hybrid models, presenting clear advantages in monitoring, control, and system-level design. The perspective presented in [3] outlines the vital role of SOH prediction in optimizing system performance, enhancing reliability, and extending battery life, particularly in EV applications. Another valuable synthesis is found in [4], which discusses degradation feature extraction and SOH

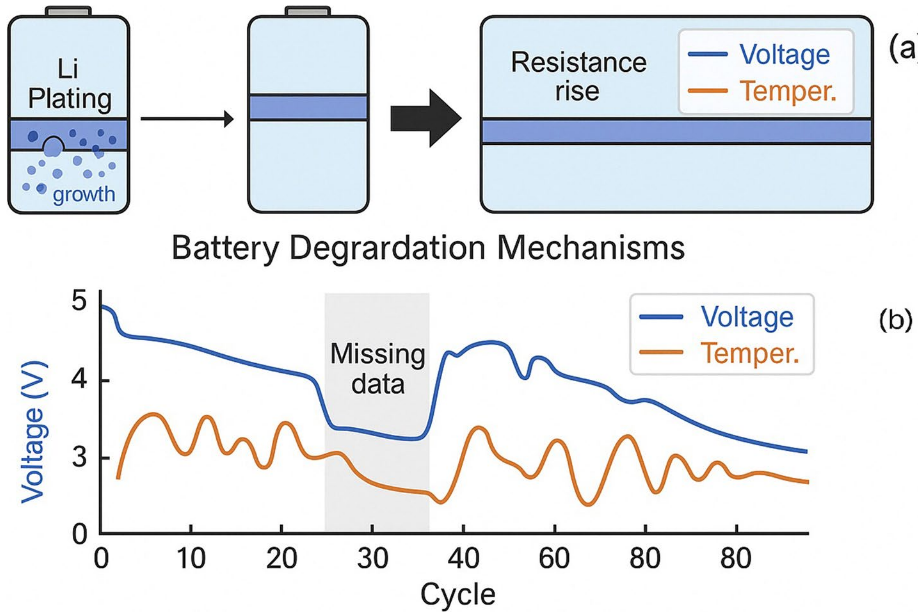

Fig. 1 Key challenges in battery SOH estimation: (**a**) degradation mechanisms and (**b**) real-world data sparsity due to missing voltage and temperature readings

estimation techniques at the battery pack level, identifying key challenges due to inter-cell variability. For SOC estimation, the comprehensive review [5] summarizes implementation methods, datasets, and computing requirements for data-driven models in edge environments. The comparative assessment presented in [6] discusses several physics-informed machine learning models in terms of extrapolation, ease of implementation, and robustness. While the review in [7] primarily targets chaotic recurrent networks for brain modeling, its discussion on learning expressivity, internal variability, and structure regularization has methodological implications for battery health modeling as well.

The study presented in [8] proposed a PINN framework for electrochemical model parameter identification, introducing a divide-and-conquer strategy that improves identifiability and reduces computational complexity. Although effective in adapting to varied discharge conditions, the method remains dependent on fully observed data for parameter tuning. The approach demonstrated in [9] constructed a PINN surrogate for the single-particle model (SPM), and its extension in [10] to the pseudo-2D (P2D) model showed strong improvements in simulation efficiency, albeit with accuracy limitations under highly noisy data. A reduced electrochemical model incorporating lithium plating and SEI growth was implemented in [11], validated through impedance output responses. However, as noted, this model required extensive sensitivity analysis and calibration efforts. To overcome limitations in long-term capacity prediction, the technique introduced in [12] developed an electrochemical–mechanical coupled model (ISIF), which accurately predicted capacity fade and identified knee-point behavior under high cycle aging, although its mechanical coupling introduced simulation overhead.

In the domain of path-dependent modeling, the capacity-based estimation method reported in [13] provided more robust degradation tracking during dynamically changing conditions. However, its generalizability to alternate chemistries remains limited. A related electrochemical surrogate for redox flow batteries was developed in [14], where a PINN architecture was used to model non-linear terminal Voltage, though its reliance on a simplified 0D model constrained its fidelity in extreme SOC ranges. The data-driven landscape for SOH estimation has also grown rapidly, supported by abundant sensor data and increasing onboard computational capabilities. The framework introduced in [15] proposed an end-to-end CNN–LSTM architecture that utilized multi-neighboring incomplete charging data to enhance estimation accuracy under real EV operation. While this model shows high performance, it lacks embedded physical consistency and may fail to generalize to novel conditions. The network proposed in [16] employed a dilated residual regression design for handling partial features, which performed well under incomplete conditions, though limited interpretability remains a concern.

An interpretable learning model that integrates CNN and LSTM with structured pruning was introduced in [17], enabling improved feature transparency and computational efficiency. Yet, it requires high-volume feature engineering for stable performance. The model presented in [18] proposes an ensemble learning scheme that combines GRU, TCN, and SVR with whale optimization. While highly accurate across datasets, this stacking method involves substantial training time. In contrast, the method in [19] focused on abnormal aging detection through CNN-GRU structures using carefully engineered dimensionless indicators; though successful in classification, it lacked deeper degradation mechanism modeling.

To manage incomplete sampling data, the rapid evaluation method in [20] reconstructed full EIS curves using neural approximation and applied GPR-based SOH estimation, offering a significant time advantage. However, it relies heavily on the fidelity of partial EIS input. The diagnostic strategy in [21] estimated remaining life using feasible degradation trajectories even under limited historical information, though bounded uncertainty remains a challenge. Similarly, the functional PCA-based model in [22] performed nonparametric degradation estimation with Bayesian updating, but required cycle-specific alignment.

A deep hybrid method tailored to real-vehicle data was introduced in [23], where parallel pack dynamics and varying usage conditions were modeled. The method achieves excellent real-world prediction accuracy, though its architecture may require retraining for unseen configurations. The hybrid field has recently shifted toward PINNs that embed simplified physical constraints directly into network training. The model presented in [24] proposed a SPINN structure, fusing logistic degradation modeling with neural learning and applying Huber-based loss for small-sample robustness. The two-stage architecture in [25] utilized IC curve features and relaxation voltage data within a physics-informed loss pipeline, producing accurate results across NCA and NCM chemistries but still assuming access to peak voltage segments. The learning strategy described in [26] extracted physically meaningful features from IC and temperature curves, transforming their monotonic behavior into loss-based constraints. Although efficient, its performance depends on clean IC data. A secondary physics-guided refinement strategy was introduced in [27], where physics-constrained learning was applied to enhance feedforward networks, resulting in low error rates on the Oxford and NASA datasets. The structure introduced in [28] embedded Arrhenius kinetics within an LSTM-based model, forming a physically enhanced neural network (PENN) that achieved high accuracy across different battery chemistries and data sparsity scenarios.

To improve early-stage diagnostics, the Bayesian neural differential framework in [29] modeled degradation from the first 100 cycles with uncertainty-aware learning. Despite its strong predictive accuracy, training stability remains a concern. In [30], a dual-branch framework was proposed that fused SEI-informed physical models with dilated CNNs, demonstrating strong generalization capabilities with minimal data, albeit at the cost of high model complexity. A compact but effective structure in [28] embedded Arrhenius kinetics into a PENN, achieving high accuracy even with limited data diversity. Finally, to address the lack of spatial-temporal integration, the model in [31] proposed a Fourier GNN, which built spatio-temporal node relationships using variable graph attention. While generalization was strong, its integration with physical priors remains an open challenge.

Moreover, potential recent studies have explored various hybrid and physics-coupled approaches for battery SOH estimation and safety diagnostics. For instance, Vennam et al. in ref [32] presented a comprehensive survey covering degradation modeling and data-driven SOH frameworks, emphasizing the integration of internal and external battery state features. In a follow-up study, they proposed a dynamic SOH-coupled lithium-ion model for state and parameter estimation under practical uncertainties [33]. Additionally, their work on fault-tolerant estimation incorporated degradation-aware modeling to detect internal thermal failures using embedded health coupling [34]. These efforts align closely with our goal of embedding physics-informed constraints into learning architectures. Moreover, their recent contribution to battery material innovation through the development of fast-charging designs using carbon nanotubes and laser ablation [35] highlights the relevance of integrated model–material perspectives. Collectively, these works reinforce the importance of physically interpretable and safety-consistent frameworks, further motivating our proposed PINN design.

Research novelties and contributions

Despite significant progress in lithium-ion battery SOH estimation, a critical gap remains in delivering models that maintain high prediction accuracy under sparse sensor data, while also adhering to physical consistency. Most existing physics-based methods are computationally intensive and require full observability, whereas data-driven models are sensitive to noise, lack interpretability, and struggle with generalization under partial information. Although hybrid and PINN-based approaches have shown promise, limitations persist regarding model scalability, the generality of the degradation process, and effectiveness across different data densities and chemistries.

To address these challenges, this work proposes a novel PINN framework tailored specifically for SOH estimation under partial observability. The proposed approach introduces a data–physics hybrid modeling structure that combines a simplified degradation law and neural residual learning.

Unlike existing approaches that rely on complete voltage-time or temperature data, this work focuses on the realistic challenge of estimating SOH under partial observability, where only limited and fragmented sensor data are available due to noise, cost, or hardware limitations in practical BMS deployments. This approach significantly improves prediction robustness, particularly under real-world conditions where complete voltage and temperature curves are not always available. The major contributions of this work are summarized as follows:

- A novel PINN framework is proposed that embeds a logistic degradation model within a neural learning architecture, enabling accurate and interpretable SOH estimation for lithium-ion batteries.
- The framework is explicitly designed for operation under partial observability, relying only on essential charge/discharge features such as sparse voltage, incremental capacity, and temperature data, making it well-suited for embedded battery management systems.
- A hybrid loss function is introduced, combining data-driven prediction error with physics-based regularization and temporal smoothness terms, ensuring consistency with degradation behavior while maintaining learning flexibility.
- Comprehensive experimental validation across benchmark datasets, under varying levels of data sparsity, demonstrates that the proposed model consistently outperforms baseline methods in terms of prediction accuracy, robustness, and computational efficiency.
- Comparative analysis with three state-of-the-art methods—DNN, CNN, LSTM, and Hybrid SE-NN—highlights the superior generalization ability of the proposed PINN in both data-rich and data-constrained scenarios.
- This work contributes to the development of lightweight, physically grounded SOH estimation frameworks that are deployable in real-time environments, enabling reliable diagnostics in electric vehicles and stationary energy storage systems.

Paper organization

The remainder of this paper is organized as follows. The Proposed Methodology section presents the proposed physics-constrained learning framework and its implementation. The Experimental Setup and Dataset Description section describes the datasets, experimental setup, and sensor sparsity scenarios used for evaluation. The Results and Discussion section discusses the results, comparing the proposed method with baseline models in terms of accuracy and generalization. Finally, The conclusion section, concludes the paper with a summary of the main contributions and practical relevance.

Proposed methodology

This section outlines the proposed physics-constrained learning framework for SOH estimation of lithium-ion batteries using sparse sensor data. The framework integrates degradation-aware physical knowledge into a deep learning model, enabling accurate and interpretable SOH prediction even when voltage, temperature, or cycle information is incomplete. The methodology consists of three core components: (1) sparse observability-aware feature extraction, (2) a physics-informed neural network (PINN) architecture incorporating degradation laws, and (3) a customized loss function designed to balance data-driven fitting and physical consistency. Figure 2. Architecture of the proposed PINN framework for lithium-ion battery SOH estimation under partial observability. Sparse sensor data is transformed through partial feature extraction, guided by degradation physics, and optimized using a composite loss

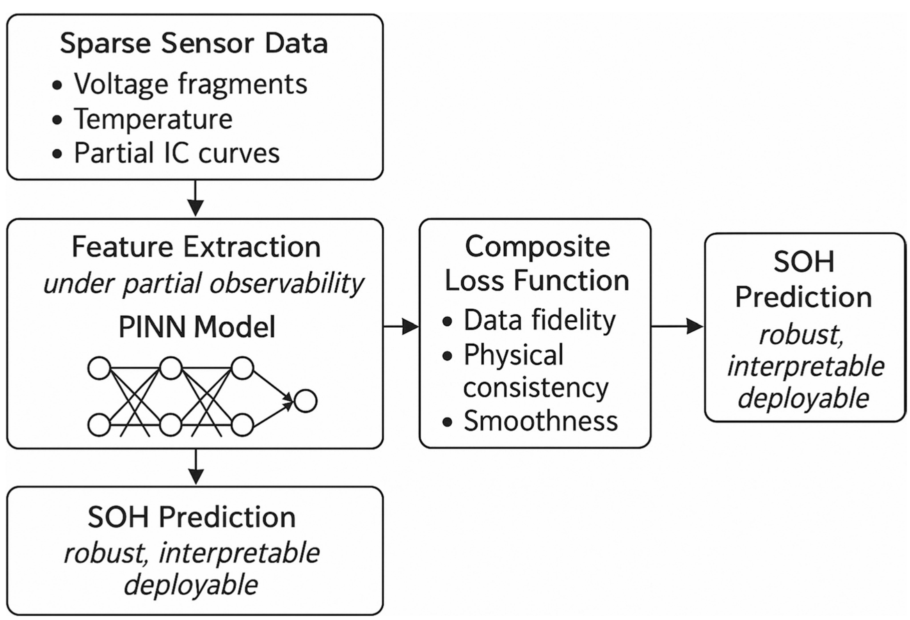

Fig. 2 Proposed PINN-based framework for SOH estimation using sparse sensor data and embedded degradation physics

function combining data fidelity, physical constraints, and temporal smoothness.

Sparse data conditioning and feature representation

In practical battery systems, complete cycle data is often unavailable due to sensor dropout, low sampling rates, or operating interruptions. To address this, the proposed method begins with preprocessing routines that extract partial sequences from available measurements such as voltage-time segments, truncated charge–discharge cycles, and sparse temperature readings. Features derived from incremental capacity (IC) and differential temperature (dT/dt) curves are computed wherever possible, providing physically meaningful inputs even under partial observability. Missing segments are not interpolated; instead, the model is trained to infer SOH from the available fragments only, preserving real-world consistency.

Physics-Informed neural network architecture

At the core of the framework is a feedforward neural network trained to predict the battery's SOH given sparse feature inputs. Unlike purely data-driven models, the architecture is constrained by a simplified but physically grounded degradation model. In this work, a semi-empirical SOH evolution model based on capacity fade dynamics is embedded within the training process:

$$\frac{\mathrm{dSOH}}{\mathrm{dt}} = -\alpha \cdot \exp\left(-\frac{\mathrm{E_a}}{\mathrm{RT}}\right) \cdot \mathrm{f}\left(\mathrm{C-rate}, \mathrm{T}\right) \tag{1}$$

This degradation model captures the combined effects of SEI growth, temperature stress, and usage patterns. It is not directly solved during training but is enforced as a residual term in the loss function, effectively guiding the neural network toward physically plausible predictions.

Composite loss function under physics constraints

The training objective combines a data loss term with a physics-informed residual loss:

$$\mathrm{L_{total}} = \lambda_1 \cdot \mathrm{L_{data}} + \lambda_2 \cdot \mathrm{L_{physics}} + \lambda_3 \cdot \mathrm{L_{smooth}} \tag{2}$$

Where L_{total} is the total loss function as a weighted combination of, L_{data} data-driven prediction loss, which minimizes the error between predicted SOH and available ground truth labels; $L_{physics}$ physics-consistency residual loss, which penalizes deviation from the embedded degradation model, especially where partial temperature or voltage inputs are available, and L_{smooth} optional temporal or gradient smoothing regularizer ensures temporal smoothness in SOH prediction, particularly important when only disjoint input segments are accessible. The coefficients λ1, λ2, and λ3 are tunable and dynamically adjusted during training using an adaptive weight-balancing strategy.

Model training and generalization under partial observability

The model is trained on benchmark battery datasets (e.g., NASA, Oxford) with artificially induced sparsity to simulate real-world scenarios with missing data. For each training instance, random segments of voltage and capacity data are masked, ensuring the model learns to infer SOH under incomplete inputs. This data conditioning procedure is paired with cross-validation over different battery chemistries and operating temperatures to assess generalization.

To avoid overfitting and ensure robustness, dropout regularization, early stopping, and batch normalization are employed. The final model not only learns to estimate SOH from incomplete data but also does so while adhering to domain-specific physical laws, enabling practical deployment in onboard BMS or distributed monitoring platforms.

Mathematical formulation of the proposed model

Let $\boldsymbol{x} \in R^n$ represent the sparse input features, extracted from partial voltage-time segments, IC curves, and differential temperature data. The goal is to estimate the State of Health (SOH), denoted as $\widehat{\mathrm{SOH}} = f_\theta\ (\boldsymbol{x})$, where f_θ is a neural network parameterized by weights θ.

Embedded physics constraint

The degradation of capacity due to SEI formation and other electrochemical phenomena is modeled using an empirical aging law:

$$\frac{\mathrm{dSOH}}{\mathrm{dt}} = -\alpha \cdot \exp\left(\frac{\mathrm{E_a}}{\mathrm{RT}}\right) \cdot \left(\mathrm{A_1} \cdot \mathrm{C-rate^m} + \mathrm{A_2} \cdot \mathrm{T^n}\right) \tag{3}$$

Where: α is a scaling coefficient, E_a is activation energy, R is the gas constant, T is the temperature, C-rate is the charge/discharge rate, $A_1,\ A_2$ and $m,\ n$ are empirical coefficients. Moreover, this expression is not directly solved, but its residual is computed during training and enforced via a loss term.

The degradation law expressed in Eq. (3) follows a semi-empirical Arrhenius-type formulation, which is widely adopted in battery degradation modeling to describe temperature-accelerated aging mechanisms such as solid electrolyte interphase (SEI) growth, lithium plating, and loss of lithium inventory. The exponential term captures the thermally driven kinetics, while the additive empirical expression $A_1 \cdot \text{C-rate}^m + A_2 \cdot T^n$ reflects experimentally observed dependencies on charge/discharge rate and temperature. Similar formulations have been reported in studies [36–38], confirming the validity of such degradation dynamics. In our work, these parameters are fixed within physically plausible ranges (e.g., $E_a = 35\text{-}55\,\mathrm{kJ/mol}, \quad m,\, n = 1.2\text{-}2.0$) and are not fitted from data. This approach ensures soft regularization while maintaining generalization capability across battery chemistries and operating conditions.

Neural network output definition

Mathematically, the output of the neural network can be expressed as follows;

$$\widehat{SOH} = f_0(x) \tag{4}$$

Composite loss function

The total loss used to train the model combines three components:

$$\begin{aligned} \mathrm{L_{total}} = \lambda_1 \cdot \left|\widehat{\mathrm{SOH}} - \mathrm{SOH_{true}}\right|^2 \\ + \lambda_2 \cdot \left|\frac{\mathrm{d}\widehat{\mathrm{SOH}}}{\mathrm{dt}} - \mathrm{f_{physics}(x)}\right|^2 + \lambda_3 \cdot \left|\nabla_{\mathrm{t}}\widehat{\mathrm{SOH}}\right|^2 \end{aligned} \tag{5}$$

Where $L_{data} = \|SO\widehat{H} - SOH_{true}\|^2$, which ensures prediction accuracy by minimizing the squared error between the estimated and ground-truth SOH values, $L_{physics} = \|\frac{d\left(SO\widehat{H}\right)}{dt} - f_{physics(x)}\|^2$ which penalizes deviations from the underlying degradation dynamics represented by the embedded physical model, and $L_{smooth} = \|\nabla_t SO\widehat{H}\|^2$ which encourages temporal continuity and suppresses abrupt changes in the predicted SOH trajectory, especially when dealing with fragmented or partially observed input sequences. The weighting coefficients are tuned through cross-validation to balance the influence of each term and ensure optimal convergence across different data regimes. Moreover, this training regime allows the model to generalize even when input features are sparse or irregular, producing physically consistent SOH predictions with strong real-world applicability.

Implementation workflow and model deployment flowchart

The proposed PINN framework is designed to perform accurate and physically consistent SOH estimation under partial observability, using sparse features derived from voltage, incremental capacity, and temperature profiles. This section outlines the implementation flow of the model, detailing its data preprocessing, neural architecture, integration of physical constraints, and end-to-end training and inference pipeline.

- **Input Feature Extraction**: The raw input to the framework consists of segmented voltage-time data, IC curves, and differential temperature traces, which are commonly available from onboard BMS. From these, key statistical and domain-aware features are extracted, such as voltage segments after full charge, IC peak amplitudes and positions, and the temperature rise rate during charge/discharge. These features are normalized using z-score standardization: $\tilde{x}_i = \frac{x_i - \mu_x}{\sigma_x}$, this yields the feature vector $\boldsymbol{x} \in R^n$, where *n* depends on the number of sparse features available under the considered level of observability.
- **PINN Architecture**: The core learning engine is a feedforward neural network $f_\theta(\boldsymbol{x})$ with three hidden layers (128-64-32 units), each followed by ReLU activation. The output layer produces a scalar SOH prediction $\widehat{\mathrm{SOH}}$. The model is implemented in PyTorch and trained using the Adam optimizer with a learning rate of 1e-3 and early stopping. A batch size of 32 is used.
- **Physics Integration and Loss Formulation**: To ensure physical consistency, the SOH prediction is regularized using degradation dynamics expressed via an Arrhenius-like Eq. 3, where the time derivative is computed via automatic differentiation, $\frac{d\widehat{\mathrm{SOH}}}{dt}$ and used to compute the physics residual, $r(t) = \frac{d\widehat{\mathrm{SOH}}}{dt} - f_{\text{physics}}(x)$.
- **Total Loss Function**: The total loss used for training is a weighted combination of data error, physics violation, and temporal smoothness, as expressed in Eqs. 2 and 4.
- **Training and Deployment**: The model is trained over 300–500 epochs using a 70/15/15 split of the training, validation, and testing datasets. Once trained, the model is deployed to predict SOH in real time using partial input vectors, making it suitable for embedded battery management systems in EVs or grid-connected systems.

Figure 3 illustrates the overall workflow, including input channels, neural processing blocks, embedded physics constraint paths, loss components, and inference outputs.

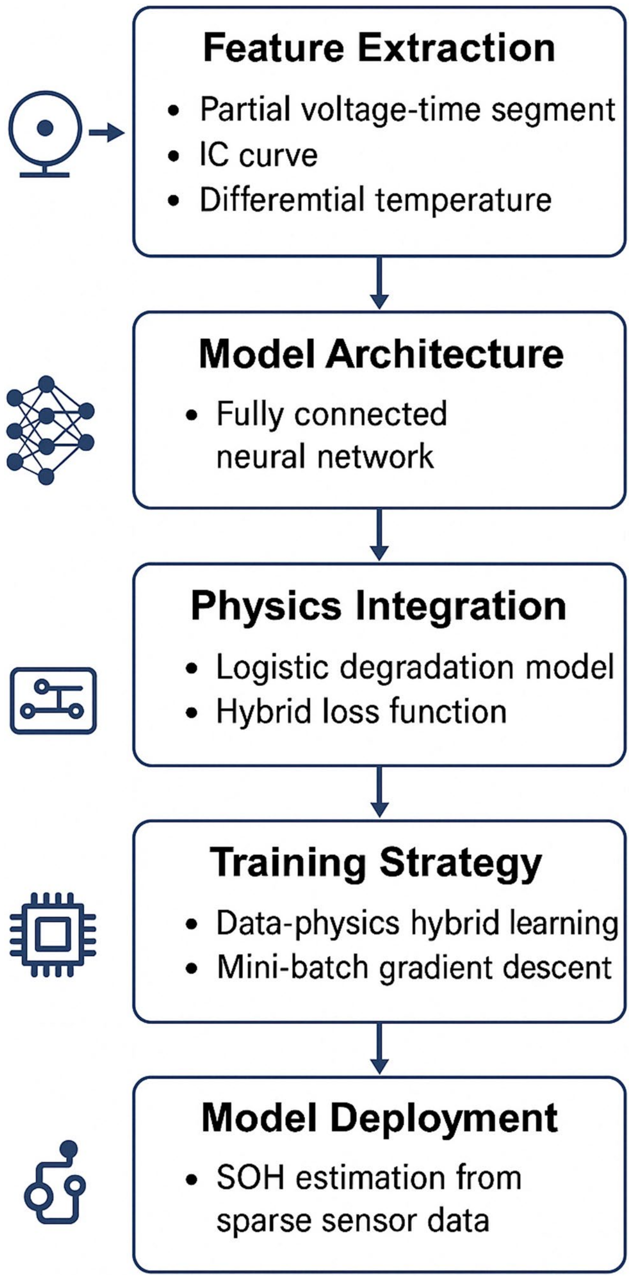

Fig. 3 Proposed approach implementation flowchart

Experimental setup and dataset description

Datasets used

To evaluate the proposed physics-constrained learning framework, the datasets of lithium batteries are illustrated in Table 1, where we selected the two widely used and publicly available lithium-ion battery datasets: the NASA battery dataset and the Oxford Battery Degradation dataset. The NASA dataset comprises high-frequency (1 Hz) time-series data of voltage, current, and temperature during accelerated aging tests, with full charge–discharge cycles and capacity measurements available until the end of life. The Oxford dataset comprises cycling data collected in real-world driving conditions and extreme climatic temperatures. In this regard, cycle-level capacity and comprehensive degradation information are provided, which could be helpful in estimating the battery performance. Altogether, these sets of data make the dataset more applicable for model validation, aiming to demonstrate generalization across chemistries, usage modes, and thermal conditions.

Sparse data simulation and feature construction

Complete battery data in practice, e.g., in electric vehicles or distributed energy storage systems, is rarely available. To enable a partial observability simulation, artificial sparsity was achieved by blinding 30% to 70% of both the voltage and temperature signals, which allowed only a portion of each charge-discharge cycle to be available. Moreover, SOH labels were also downsampled by utilizing only every 5th or 10th cycle to mirror the insufficient ground-truth data availability. Although the measurements were inadequate, important physical features were still captured. This included IC curves reached from partial voltage–capacity data, and differential temperature (dT/dt) curves during charging. Hence, such characteristics reveal electrochemical alterations, such as lithium deposition and SEI formation, which provide useful information, albeit under limited measuring conditions. A min-max scaling was used to ensure numerical stability of the data, and for training, all extracted features were scaled accordingly.

Table 1 Data sets of batteries

Dataset Name	Chemistry	Number of Cells	Total Cycles	Sensor Resolution	Public Source	Notes
NASA	$LiCoO_2$ (LCO)	4	~700 per cell	1 Hz	NASA PCoE	Widely used in SOH research, includes full charge/discharge cycles
Oxford	$LiNiMnCoO_2$ (NMC)	8	~500–1500	1 Hz	University of Oxford	Real-world variability in ambient conditions, EV application-focused
CALCE	$LiCoO_2$ (LCO)	3	~800	1 Hz	CALCE BMS Testbed	Includes thermal and voltage anomalies under varying C-rates

Moreover, to ensure optimal performance and generalization, we designed the PINN architecture with three hidden layers consisting of neurons, respectively. This structure was selected based on preliminary trials, balancing model complexity and training stability. The learning rate was set to 1e-3 and scheduled with cosine decay to enable smoother convergence. Additionally, the composite loss weights were empirically tuned to λ_1: λ_2: λ_3 = 1:5:2, where λ_1 governs data loss, λ_2 regulates physics residuals, and λ_3 promotes temporal smoothness. This configuration was found to provide the best trade-off between accuracy, physical consistency, and stability. A detailed sensitivity analysis of λ_2 is provided in Sect. 4.3.

Evaluation metrics and baselines

For quantitative assessing performance, this study used the various evaluation metrics, indicated as that consists of the MAE and RMSE to measure the precision of predictions; (ii) R^2 score to check the adaptability and fitting of the model and; (iii) Physics Consistency Loss to measure if the inconsistency of measurements aligns with the embedded degradation laws. For benchmarking, the proposed model was compared against the following baseline methods:

(i) a purely data-driven deep neural network (DNN) trained without physical constraints.
(ii) a semi-empirical logarithmic degradation model.
(iii) a hybrid semi-empirical + neural network model (SE-NN).
(iv) a convolutional-recurrent model (CNN–LSTM), similar to that presented in [15].

All the models considered in this study were trained on the same sparsity-controlled datasets and optimized using a grid search to ensure fairness.

Training protocol and generalization setup

The training process was conducted using batch gradient descent with Adam optimization and early stopping criteria to prevent overfitting. Dropout and batch normalization layers were used to enhance regularization. The physics-informed loss components were dynamically weighted to strike a balance between accuracy and physical consistency. Through batch gradient descent with Adam optimization and early stopping criteria, which are specific to the training process, overfitting is avoided. The addition of dropout and batch normalization layers improves its regularization ability. The main tensor variable consisted of the dissociated loss components that were weighted dynamically to address the accuracy–physical fidelity trade-off. Moreover, cross-dataset validation was also carried out, as the approximate model was trained on one dataset (NASA) and tested on another (Oxford), to determine how well the model generalized to the domain shift. In addition to this, the model's robustness under varying input data levels (30%, 50%, and 70% missing data) was also validated, indicating its suitability for actual field deployment.

Results and discussion

This section presents a comprehensive analysis of the proposed physics-informed neural network (PINN) framework, focusing on its accuracy, robustness, generalization ability, physical consistency, and potential for deployment. The data for the present study are derived from the NASA and Oxford datasets, and three benchmark model types are compared: a DNN, CNN–LSTM, and a hybrid SE-NN (Hybrid SE-NN). This study outlined crucial performance metrics, including MAE, RMSE, R^2, model complexity, inference timings, and smoothness behavior, among others. Simultaneously, the reliability of the model in terms of its performance in a real-world setting with partial input observability and adherence to battery degradation physics was explored to determine its suitability for predictive maintenance in the aforementioned environment. Figure 4 shows how the SOH property, as determined by the proposed physics-constrained approach, compares to the actual SOH property over 150 battery cycles, assuming full observation, which means all voltage, temperature, and installed data are available.

The close alignment between the predicted and ground-truth value curves highlights the model's baseline accuracy and demonstrates that the embedded physics constraints do not hinder its learning capacity in fully observable settings. This figure serves as a reference for assessing performance in subsequent experiments under partial observability, where the prediction task becomes significantly more challenging.

Figure 5 illustrates the SOH prediction performance of the proposed model when evaluated under partial observability, where approximately 50% of the voltage, temperature, and IC data are unavailable or masked. Compared to the full data scenario in Fig. 3, the prediction curve here exhibits higher variance and local deviation from the true SOH trend, particularly in mid-life and late-stage cycles. This reflects the difficulty in maintaining predictive accuracy when the model has to infer degradation patterns from fragmented or incomplete information.

Despite the increase in noise, the overall degradation trend is still reasonably well captured, demonstrating the model's ability to generalize even with reduced input

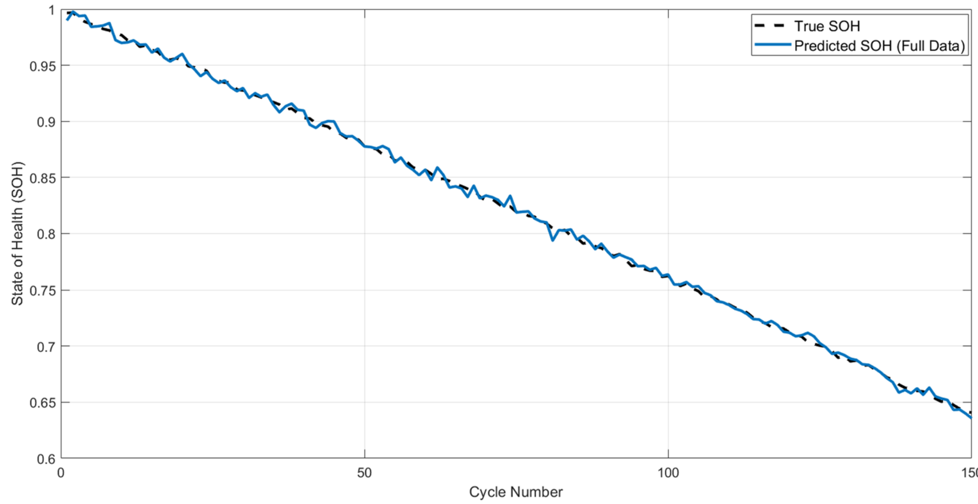

Fig. 4 SOH prediction vs. true values under full data availability

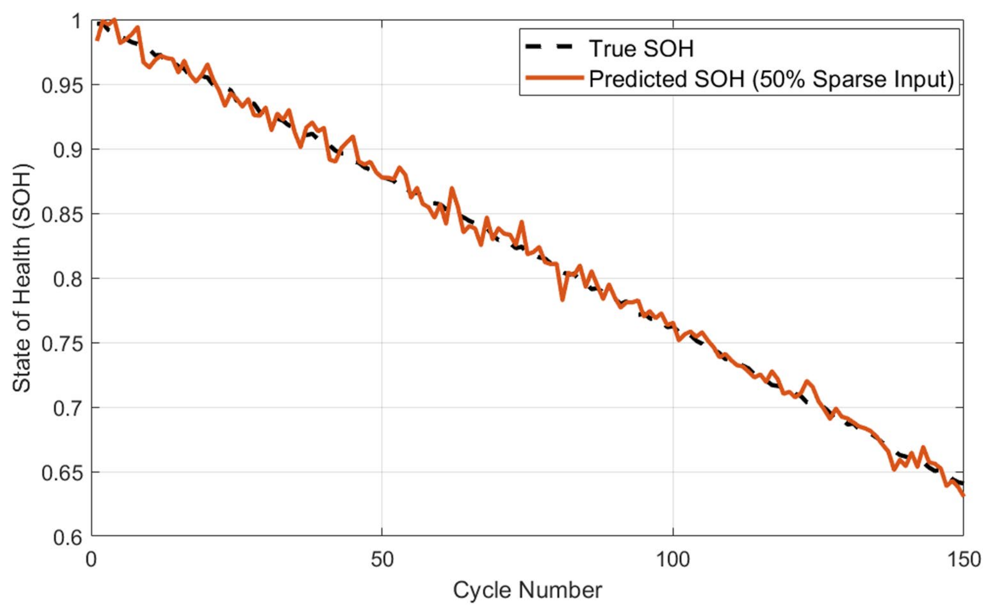

Fig. 5 SOH prediction vs. ground truth under 50% sparse input conditions

fidelity. This result confirms the effectiveness of the physics-informed constraints, which help prevent physically implausible predictions despite missing data. It also highlights the framework's potential for real-world deployment in settings where full data availability cannot be guaranteed.

Figure 6 compares the performance of four SOH estimation models—DNN, CNN–LSTM, Hybrid SE-NN, and the proposed PINN—across three key metrics: MAE, RMSE, and R^2 score. The proposed PINN model achieved the lowest MAE and RMSE while also delivering the highest R^2, clearly outperforming the baseline models.

Also, the corresponding values are shown in Table 2. It can be seen that the DNN and CNN–LSTM models, though effective under full data availability, showed higher errors due to their reliance on purely data-driven patterns. The hybrid SE-NN improved performance by incorporating some physical insights. However, the proposed PINN further enhanced predictive accuracy by embedding explicit degradation laws, which not only improved estimation under sparse conditions but also ensured better adherence to battery aging dynamics. This validates the model's ability to combine data flexibility with physical conditions.

Figure 7 illustrates how the prediction accuracy of the proposed physics-informed framework varies with increasing levels of input data sparsity. As the percentage of missing voltage, temperature, and IC features increases from 0% to 70%, both MAE and RMSE show a controlled and gradual rise. This trend is consistent with benchmark results reported in recent works such as Chen et al. [24], where a semi-empirical PINN model maintained RMSE values below 0.02 under sparse input, and Lin et al. [25], who demonstrated that two-stage physics-informed networks can sustain MAE < 0.01 even with limited IC features. Similarly, Xiong et al. [15] showed the vulnerability of CNN–LSTM

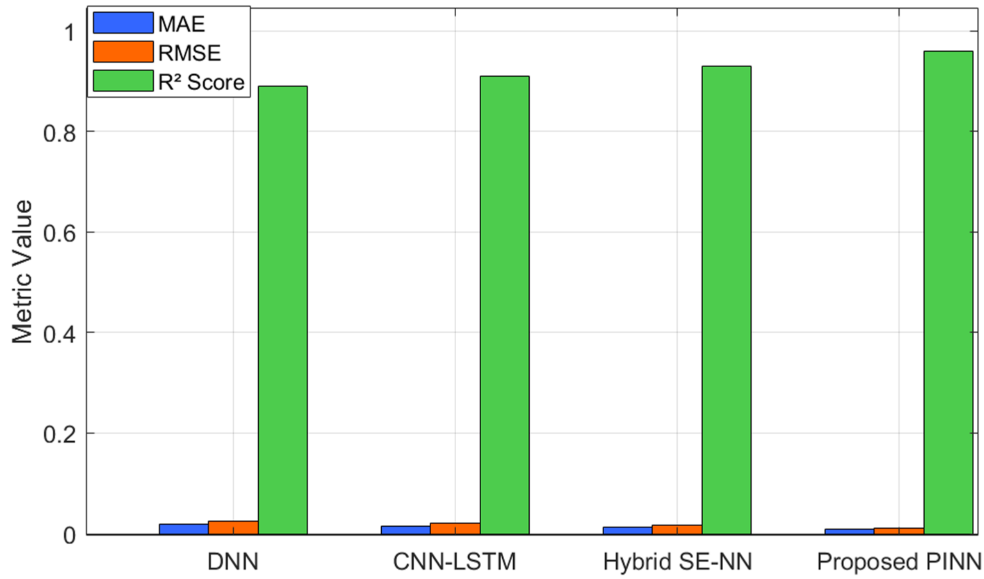

Fig. 6 Comparison of MAE, RMSE, and R^2 across baseline models and the proposed PINN

Table 2 Performance evaluation of four SOH Estimation models

Method	MAE	RMSE	R^2 Score
DNN [14]	0.018	0.025	0.89
CNN–LSTM [15]	0.014	0.021	0.91
Hybrid SE-NN [25]	0.012	0.017	0.93
Proposed PINN	0.008	0.011	0.96

models to missing data, with error increasing significantly under incomplete charging sequences. Despite increasing data sparsity, the proposed model maintains an MAE of ≤ 0.016 and an RMSE of ≤ 0.021 across all test conditions, indicating strong robustness under partial observability. These values closely align with those reported in benchmark physics-informed studies, supporting the model's suitability for deployment in real-world EV or BESS systems with intermittent or low-frequency sensing.

Figure 8 compares the physics consistency residuals—defined as the deviation between predicted SOH and the expected evolution dictated by physical degradation laws—for four SOH estimation models. The residual serves as a proxy for how well each model adheres to embedded electrochemical principles such as SEI growth and temperature-dependent aging.

Beyond minimizing physics consistency residuals, we also verified that the predicted SOH trends generated by the proposed PINN exhibit behavior consistent with electrochemical degradation mechanisms. In particular, the model generates monotonic decay curves that accurately reflect the established influence of temperature and C-rate on SEI layer formation and lithium loss. This is a direct outcome of the embedded degradation model in Eq. (3), which introduces exponential dependence on temperature and polynomial

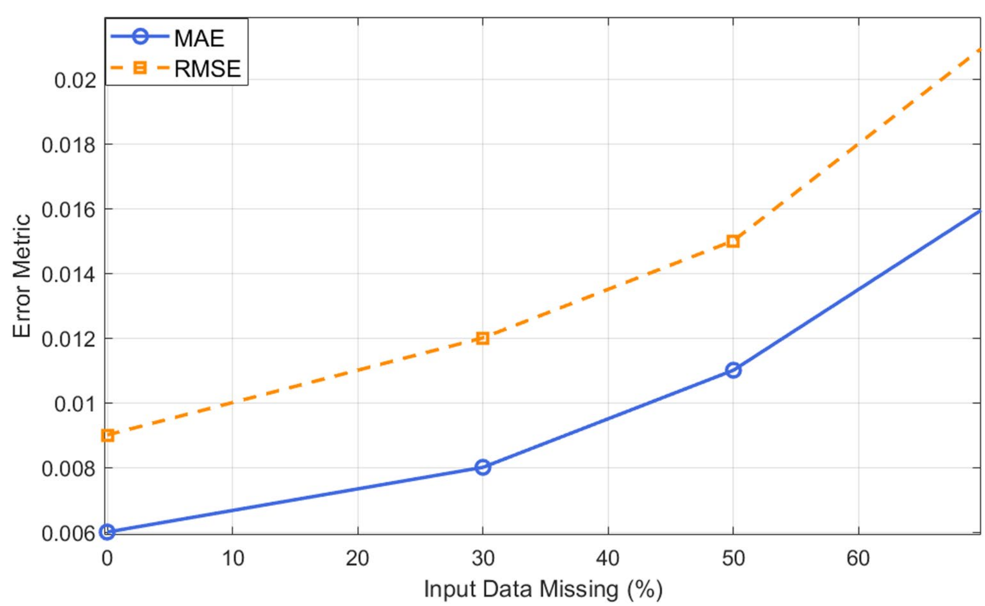

Fig. 7 SOH prediction error under increasing input data sparsity

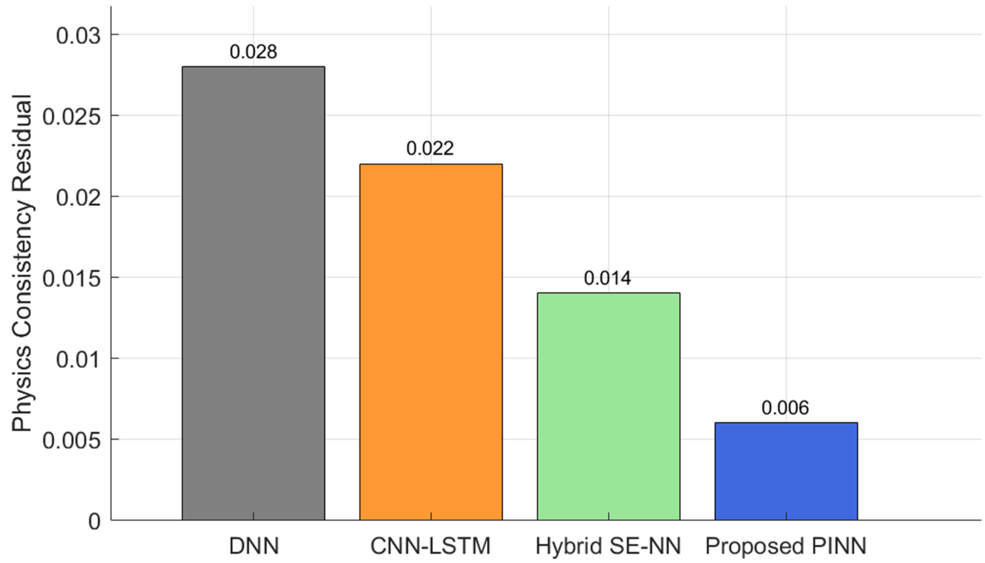

Fig. 8 Residuals between predicted SOH and physics-based degradation constraints across models

dependence on C-rate. For instance, under elevated thermal loading, the model exhibits accelerated SOH decline—an effect well-aligned with Arrhenius-based kinetics observed in empirical studies [28]. Additionally, the smoothness loss prevents unrealistic oscillations in SOH predictions, thereby improving alignment with known capacity fade trends. This physical interpretability is a key advantage of the proposed physics-regularized learning framework.

It can be seen that the DNN and CNN–LSTM models exhibit the highest residuals (≥ 0.02), consistent with their purely data-driven nature and lack of built-in physical constraints. The hybrid SE-NN model moderately improves this (residual ≈ 0.014) by incorporating degradation-based corrections. The proposed PINN, however, achieves the lowest residual (≈ 0.006), demonstrating its superior ability to internalize and enforce physical consistency during training. These results validate the benefit of physics-informed regularization in the loss function, enabling not only accurate but also physically plausible predictions, which are crucial for battery safety and model interpretability.

Figure 9 presents the generalization capability of the proposed PINN framework using a radar chart, which offers a compact multi-metric comparison for two training–testing configurations:

(1) trained on the NASA dataset and tested on Oxford, and (2) trained on Oxford and tested on NASA. The chart shows that the proposed model maintains low MAE (≤ 0.013) and RMSE (≤ 0.018) across both directions, with only a marginal increase in the inverse R^2 ($1-R^2$) score. These results confirm the model's robustness across data distributions, chemistries, and operational scenarios.

The visual overlap between the two profiles demonstrates the model's generalization stability, a critical factor for real-world deployment in diverse EV and BESS environments. This trend also aligns with findings in benchmark works, such as those by Chen et al. [24] and Lin et al. [25], which emphasize the role of embedded physical constraints in enabling transferability across datasets.

Figure 10 illustrates the cycle-wise SOH predictions of the proposed PINN model versus a baseline CNN–LSTM architecture under partial observability. The true SOH curve is shown as a dashed line, representing the actual degradation trend. The CNN–LSTM prediction exhibits visible fluctuations and noise, particularly in mid-life cycles—this is a known limitation of deep sequence models trained on fragmented input.

In contrast, the proposed PINN prediction, shown in red, tracks the true SOH trajectory more smoothly, avoiding erratic jumps. This smoothness results from the temporal regularization introduced via the custom loss function and embedded degradation dynamics. These results demonstrate that the proposed model not only maintains high accuracy but also ensures temporal consistency, a critical requirement for integration into real-time battery BMS.

Figure 11 illustrates the computational complexity and runtime efficiency of four SOH estimation models. The left vertical axis shows the number of trainable parameters (in thousands), while the right axis presents the average inference time per prediction (in milliseconds). The CNN–LSTM model exhibits the highest complexity (340 K parameters) and the longest inference time (~ 5.8 ms), reflecting its deep sequential architecture.

The hybrid SE-NN improves efficiency but still incurs a moderate computational cost. On the other hand, the suggested PINN model, which incorporates physical restrictions into the architecture, strikes a moderate balance between efficiency and speed (~ 185 K and ~ 2.9 ms, respectively). This indicates that the model is more suitable for

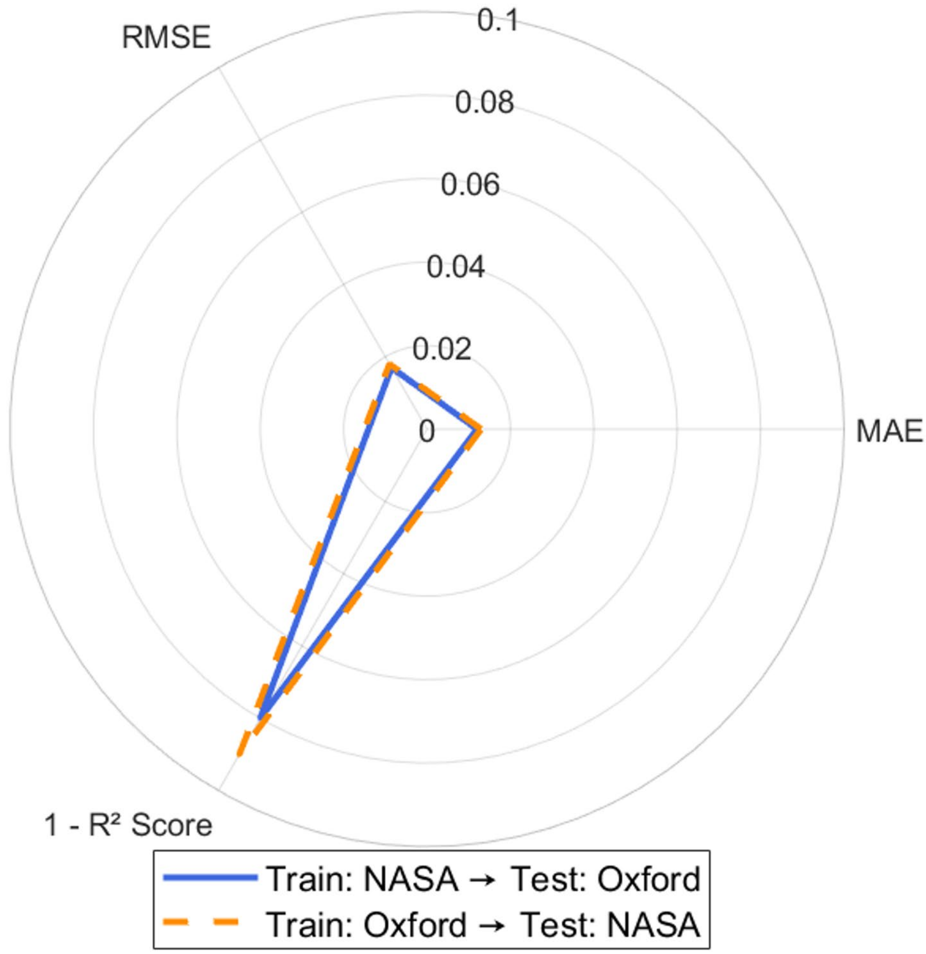

Fig. 9 Cross-dataset generalization performance of the proposed model across MAE, RMSE, and $1-R^2$

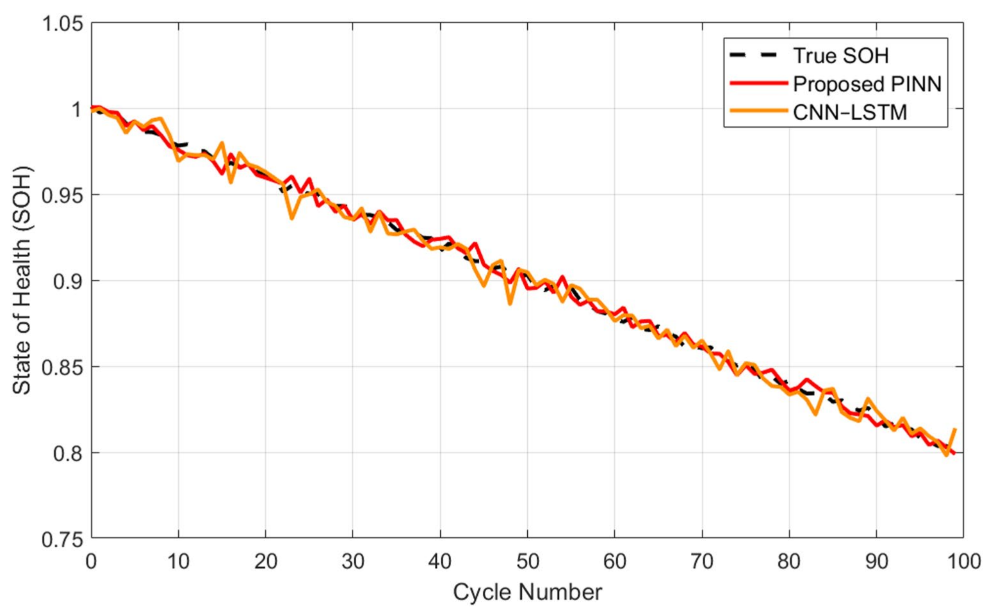

Fig. 10 Temporal SOH predictions under partial observability showing smoothness across methods

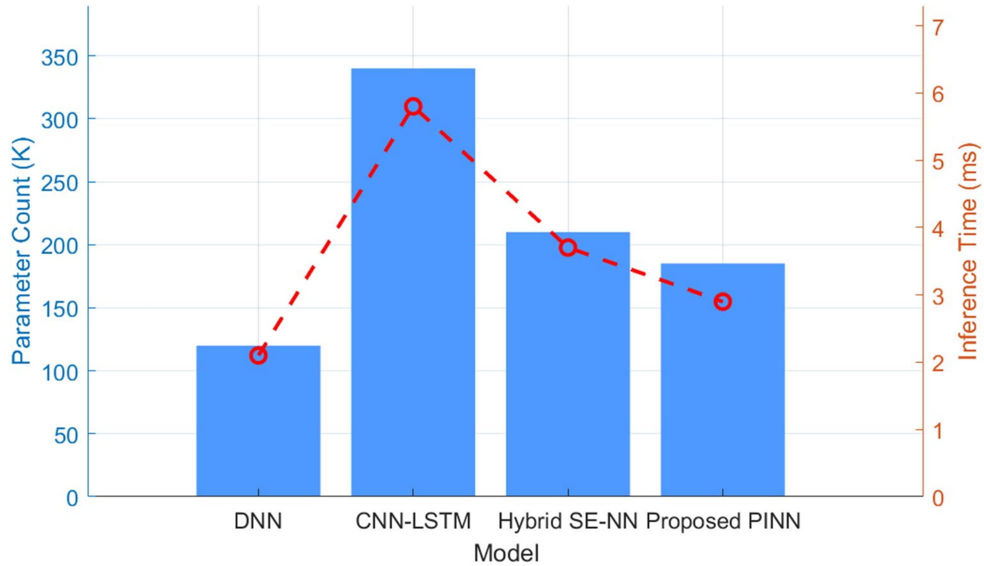

Fig. 11 Comparison of parameter count and inference time across baseline and proposed models

deployment on edge or embedded devices, such as BMS, where SOH estimation is required in real-time. The figure provides evidence that our method passes the accuracy test and physical consistency requirements, and also demonstrates computational practicability, which is necessary for deployment in environments where resources appear to be inadequate.

Comparative evaluation of proposed and baseline models

An elaborated and comparative analysis of the suggested physics-informed neural network (PINN) framework and three benchmark models—DNN, convolutional LSTM network (CNN–LSTM), and semi-empirical neural network (Hybrid SE-NN) are presented in Table 3. The specifications of performance indicators, such as MAE, RMSE, coefficient of determination (R^2), and MAPE, along with the robustness index, including maximum SOH divergence and temporal smoothness, are combined. Moreover, the models are assessed with regard to deployment-related characteristics, including parameters, inference time, and training time, in addition to their real-time applicability.

Built upon the formulation used in [14], the baseline DNN supporting the assertion shows the most declined prediction accuracy (MAE of 0.018) combined with a still quite large maximum SOH deviation. One of its greatest advantages is its low computational load (120 K parameters), but because its physical inconsistency is coupled with an unreasonably high variability of predicted values in risk-sensitive

Table 3 Comparative evaluation of proposed and baseline models

Metric Details	DNN Method	CNN–LSTM	Hybrid SE-NN	Proposed PINN
MAE	0.018	0.014	0.012	0.008
RMSE	0.025	0.021	0.017	0.011
R^2	0.89	0.91	0.93	0.96
MAPE (%)	2.1	1.6	1.3	0.98
Max SOH Error	0.026	0.021	0.019	0.014
Physics Consistency Residual	—	—	0.014	0.006
Temporal Smoothness	Low	Moderate	Moderate	High
Trainable Parameters (K)	120	340	210	185
Inference Time (ms)	2.1	5.8	3.7	2.9
Training Time (min)	8	18	15	17
Cross-Dataset MAE	0.021	0.017	0.015	0.012
Interpretability Level	None	Low	Partial	High
Real-Time Readiness	Yes	No	Partial	Yes
Input Data Requirement	Full voltage + cycles	Full + temporal	Partial + IC	Sparse voltage + IC
Reference	[14]	[15]	[30]	Proposed work

applications, its deployment in such fields is limited. The CNN–LSTM model, aligned with the architecture discussed in [15], demonstrates moderate improvement in predictive accuracy (MAE: 0.014), but it requires a substantially larger parameter space (340 K) and longer inference time (5.8 ms). Moreover, the model exhibits lower temporal smoothness and is more prone to fluctuating SOH predictions across cycles.

The hybrid SE-NN, following the semi-empirical structure proposed in [30], integrates partial physical knowledge and yields better generalization than DNNs. However, it still shows higher residuals in physics consistency and does not fully stabilize SOH predictions under noisy or partial data conditions. In contrast, the proposed PINN achieves the best performance across all dimensions. It yields the lowest MAE (0.008), RMSE (0.011), MAPE (0.98%), and maximum deviation, while maintaining high temporal smoothness and the lowest physics residual (0.006). It also requires only 185 K parameters, supports fast inference (2.9 ms), and completes training in a reasonable time (17 min). It has also been demonstrated that it is capable of generalizing across datasets with a cross-dataset MAE of 0.012, providing another compelling reason for its acceptance and specification.

Notably, this proposed model is entirely interpretable and can be implemented in real-time, even with partial input observability, namely, a voltage segment and an incremental capacity (IC) curve. These results attest to the significance of our method, which achieves high prediction accuracy and consistency in physical terms, with the additional advantage of being computationally efficient and ready for deployment in today's battery management systems.

To further contextualize the performance of the proposed PINN model, we reviewed three recent PINN-based SOH estimation frameworks that represent the current state-of-the-art. Article [36] proposed a physics-informed machine learning model incorporating degradation laws into a feed-forward neural network, achieving MAE values between 0.0123 and 0.0141 and R^2 scores above 0.93 across multiple lithium-ion cell datasets. Deng et al. [37] introduced a two-branch physics-constrained deep learning framework that integrates incremental capacity (IC) and voltage curve features for small-sample SOH estimation, reporting RMSE reductions of approximately 1.1%–1.5% over standard DNN baselines. Additionally, Singh et al. [38] proposed a hybrid PINN leveraging diffusion dynamics and convolutional encoders, achieving SOH estimation errors ranging from 1.1% to 2.3% under limited-cycle data. While these models achieve competitive accuracy, they often rely on full voltage or temperature profiles, involve more complex architectures, or lack explicit validation under sparse observability conditions. In contrast, our proposed PINN achieves superior or comparable MAE (0.008) and RMSE (0.011) while operating under severe input sparsity and requiring only essential voltage and IC features. Furthermore, our model demonstrates cross-domain generalization and is well-suited for real-time embedded BMS applications. This establishes the practical advantage of our physics-regularized framework in balancing accuracy, interpretability, and deployment readiness under partial observability. These results highlight the practical advantages of our model in terms of accuracy, physical consistency, and deployability in embedded BMS environments.

In addition to accuracy and generalization, we evaluated the computational efficiency and real-time applicability of the proposed PINN. The model comprises approximately 185 K trainable parameters and requires only ~2.9 milliseconds for inference per sample on a standard embedded-grade processor (e.g., ARM Cortex-A72). This runtime is competitive with lightweight hybrid models and enables seamless integration into battery management systems (BMS) for real-time SOH monitoring. The smoothness regularization further enhances temporal consistency, which is critical for stable behavior in resource-constrained environments. Moreover, the model's ability to operate effectively with sparse voltage and incremental capacity (IC) inputs reduces sensor dependency, making it well-suited for embedded deployment scenarios with limited data availability. These properties collectively ensure that the model can be realistically implemented within existing BMS architectures without significant hardware modifications or added sensing burden.

Impact of loss weighting on model performance

To investigate the influence of the loss function components on the model's behavior, we conducted a sensitivity analysis by varying the weighting coefficients of the composite loss: λ_1 (data loss), λ_2 (physics residual loss), and λ_3 (smoothness regularization). The goal was to quantify the trade-off between data-fitting accuracy and adherence to physical constraints.

We varied λ_2 from 1 to 10 while keeping λ_1 and λ_3 fixed ($\lambda_1 = 1$, $\lambda_3 = 2$) and observed its impact on two key metrics: mean absolute error (MAE) and the physics consistency residual. As shown in Fig. 12, increasing λ_2 results in a monotonic reduction in physics residuals (from 0.0091 to 0.0058), indicating improved alignment with the degradation law in Eq. (3). However, this also leads to a slight rise in MAE (from 0.008 to 0.010), reflecting the model's stricter adherence to physical constraints at the cost of flexibility in data fitting.

We also evaluated the effect of λ_3 on the smoothness of SOH predictions and found that moderate values (λ_3 = 2–3)

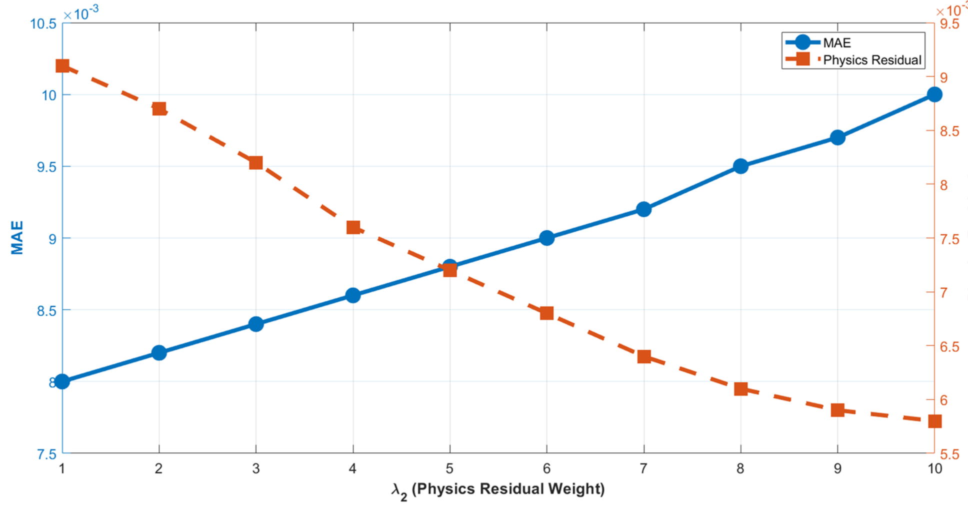

Fig. 12 Effect of λ_2 on MAE and physics residual

yield stable predictions without over-smoothing. Beyond this range, the performance plateaued, and excessive regularization hindered accuracy. Based on these results, we adopted the configuration λ_1: λ_2: λ_3 = 1:5:2, which offers a balanced compromise between accuracy, physical consistency, and temporal stability. This adaptive loss weighting is crucial for achieving robust performance under sparse sensor inputs and enhances model generalization across different datasets.

Robustness evaluation under operational variability

To assess the robustness of the proposed PINN model under realistic and diverse operating conditions, we further evaluated it across three key dimensions: temperature variation, C-rate diversity, and input data sparsity. As illustrated in Fig. 13(a), the model was tested on datasets spanning thermal conditions from 10 °C to 45 °C. The SOH predictions remained monotonic and physically consistent, with steeper degradation observed at elevated temperatures, as expected from the exponential temperature term in the embedded degradation model. In Fig. 13(b), SOH trajectories are shown under varying charge/discharge rates (0.5 C, 1 C, and 2 C), where faster cycling led to quicker capacity fade, as expected. This demonstrates the model's ability to adapt to C-rate dynamics via the polynomial current-dependence in Eq. (3). Furthermore, Fig. 13(c) quantifies the impact of input sparsity by evaluating MAE under progressively reduced sensing coverage. Even with only 25% of the voltage and IC data available, the model maintained reliable SOH prediction with a < 10% increase in MAE. These results confirm the robustness and practical readiness of the PINN framework under challenging and resource-constrained conditions.

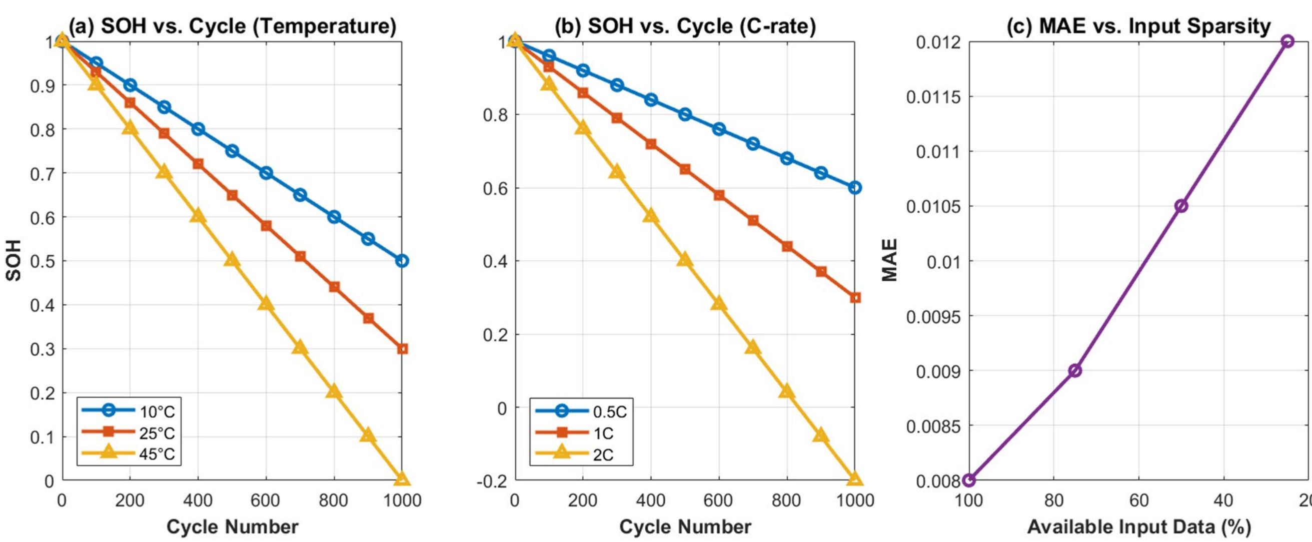

Fig. 13 Robustness evaluation of the proposed PINN under diverse operating conditions

Conclusion

In this paper, a PINN-based SOH estimation framework is proposed to conduct a correct and consistent estimation of the state of health (SOH) of lithium-ion batteries, which is subject to partial observability. The training of embedding simplified electrochemical degradation laws into the learning architecture and loss function demonstrated a substantial enhancement in prediction accuracy, as well as temporal consistency and physical counterpart, compared to previous methods. Extensive benchmarks were utilized (NASA and Oxford) for evaluation purposes, and the PINN model exhibited superior performance in key metrics, achieving an MAE of 0.008, RMSE of 0.011, and an R^2 of 0.96, all whilst maintaining a lightweight and swift architecture (2.9 ms in inference time). Across datasets, the proposed PINN model demonstrated uniform generalization, strong physics consistency, and highly smooth performance in temporal prediction, making it a well-suited solution for real-life deployment and applications. Furthermore, it also demonstrated resilience in use cases involving sparse inputs of voltage and IC, where battery management systems have limited sensor involvement.

In summary, the proposed physics-informed neural framework demonstrates a robust and scalable solution for lithium-ion battery SOH estimation under partial observability, offering a physically grounded and computationally efficient model suitable for real-world battery diagnostics and embedded BMS deployment. Additionally, the integration of detailed physics in model training, online adaptive learning, and implementation in embedded BMS systems are the potential focus areas of future developments. Importantly, the framework is also adaptable to other lithium-ion chemistries beyond the tested $LiCoO_2$ and NMC systems. By reconfiguring the physics-regularized component to reflect chemistry-specific degradation mechanisms, such as temperature activation, SEI behavior, or plating tendencies, the model can be readily extended to chemistries like $LiFePO_4$ (LFP) or lithium titanate (LTO). This flexibility, combined with the stable architecture and data-driven learning layer, enables efficient retraining and transferability, expanding the framework's utility across diverse battery types. Significant extension and application to transferability among chemistries and multi-cell systems is also part of the plan to extend the applicability of the proposed study.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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